ARTICLE

PRISM: an auditable pipeline for post-treatment glioma surveillance, evaluated branch by branch

[PENDING: A. N. Author][a] and [PENDING: B. N. Author][b]

[a][PENDING: Affiliation, address]; [b][PENDING: Affiliation, address]

ARTICLE HISTORY

Compiled October 16, 2026

ABSTRACT

PRISM predicts documented progression within 120 days of a glioma surveillance MRI and returns an audit record and a clinician report per scan. Its first version scored AUC 0.80 on one MU-Glioma-Post test split but averaged 0.72 over 500 similar splits; that split was easy because of its mix of scans before radiotherapy. Under repeated cross-validation, clinical and treatment-course variables reached 0.70 in grade 4 glioma, or 0.68 after removing leaks found by its own audit. Radiomics, frozen and fine-tuned deep encoders, resegmentation, a cleaner label and diffusion showed no measurable gain, and a locked model scored 0.61 on an external cohort. The phase of care drove discrimination, even after correcting for censoring. Reports written from the audit record by a local language model passed a faithfulness check for all 249 grade 4 scans and met every prespecified usefulness item in 247; clinicians have not yet assessed them.

KEYWORDS

glioma surveillance; prediction model evaluation; auditable decision support

1. Introduction

Most patients with high-grade glioma progress after surgery and chemoradiotherapy, and they are followed with MRI every few months. At each surveillance scan the team asks how likely it is that progression will be documented before the next decision point. Two things make this hard. Enhancement can increase after radiotherapy (RT) without tumour growth. This pseudoprogression is most common in the first 12 weeks after chemoradiotherapy and more frequent with MGMT promoter methylation (Brandes et al. 2008; Taal et al. 2008), which is why the RANO criteria ask for confirmation of progression in that window (Wen et al. 2010, 2023). And in routine data, progression is a documentation event. It depends on imaging, on symptoms and on when the next scan happens to be scheduled. A model for this endpoint is a prognostic aid for deciding whether to bring the next scan forward or act on an equivocal finding. For that use, a calibrated probability, an honest interval and a way to check what the model knew matter as much as a high AUC.

Most imaging work addresses a neighbouring question: telling pseudoprogression from true

CONTACT [PENDING: A. N. Author. Email: ...; ORCID: ...]

progression at a lesion that is already suspicious. There, machine learning models pool to an AUC of about 0.77 (Booth et al. 2022), and diffusion and perfusion measures reach 0.90 to 0.93 (Taylor et al. 2022). Forward prediction from every surveillance scan has had less attention. On the public MU-Glioma-Post dataset (Mahmoud et al. 2025), a calibrated radiomics model reported AUC 0.80 on one held-out set of patients (Christodoulou et al. 2026), and a model of volume increase at the next scan reported 0.728 under nested cross-validation (Rai 2026) [CHECK: Rai 2026 unverified in COMPARABLE_RESULTS.md].

The wider literature describes three ways such numbers mislead. Single train–test splits are unstable in small radiomics cohorts: in one brain tumour dataset, one random split tested at 0.667 and another at 0.911 (An et al. 2021). Performance drops on external data, by 0.10 or more for about a quarter of externally validated deep learning algorithms (Yu et al. 2022). And clinical comparators are rarely reported, although where they are, they hold their own. Clinical variables with MGMT reached 0.69 for pseudoprogression, and adding radiomics made the model worse (Leone et al. 2023); a clinical model reached 0.727 externally against 0.717 for imaging alone (Gomaa et al. 2025); and in survival prediction clinical variables reached 0.74 against 0.75 with radiomics added (Abler et al. 2025).

A fourth problem has had little attention. In both of the best documented clinical comparators, time since the end of RT was an input, and in one it was the most influential (Gomaa et al. 2025; Leone et al. 2023). A surveillance cohort mixes scans taken before RT, during it, in the 12 weeks after it and in later follow-up, and event rates differ sharply between these phases of care. A model that only recognises the phase can rank scans well whenever the test set holds many easy phases. Routine tables add a related trap (Kapoor and Narayanan 2023). Treatment fields are recorded once per patient, so if they are read as of the end of follow-up, a scan taken before RT "knows" about the RT that follows it.

We built PRISM (Post-treatment Response Intelligence and Surveillance Monitor) for this task. It predicts documented progression within 120 days from everything known on the scan day, explains each prediction and writes a report for the clinician. Its first version, PRISM v1, scored AUC 0.80 on one split of 30 test patients, which we call our initial evaluation split. Before building on that number we tested it, and each of the problems above turned out to apply. This paper follows that test. We show how the initial split misled and why. We then take PRISM apart branch by branch under repeated, nested cross-validation, with an external cohort and a third cohort with diffusion imaging, and find that the phase of care, not MRI texture, carries the signal. Finally we describe how PRISM was rebuilt around what remained: a modest core model whose every prediction comes with an audit record and a report that can be checked against it.

2. Methods

2.1. PRISM

PRISM takes one MRI (T1, T1c, T2 and FLAIR with a four-compartment tumour mask) and the patient's clinical tables, and returns the risk of documented progression within 120 days, an audit record and a clinician report (Figure 1). Every input is read as it stood on the scan day. Five branches can feed one ridge logistic model: (1) clinical and molecular data; (2) the treatment course, from the time since diagnosis and RT and the treatment status on the scan day; (3) imaging measurements

(volumes, morphology and change); (4) radiomics; and (5) a SwinUNETR image encoder, frozen or fine-tuned. Any branch can be switched off, which is how we ablate it. An audit layer wraps the model and writes one record per scan, and a report generator turns each record into a report (Sections 2.7 and 2.8).

PRISM v1 differed in four ways. Its radiomics branch, which we call PRISM's initial radiomics model, was a calibrated logistic regression on T1c and FLAIR radiomics plus clinical codes. Its image encoder was a small transformer of the SwinUNETR type with untrained weights. A fusion network combined the encoder output with the radiomics score. The report came from hard-coded RANO rules and a large language model (GPT-4.1-mini). When we reproduced v1, the fusion network did not beat its radiomics input (AUC 0.8006 against 0.8043), the untrained encoder carried no signal (about 0.50 alone), and the report's RANO measurements were derived from the model score rather than from the masks (Supplementary S7). We replaced the encoder with pretrained frozen encoders and the fusion network with one regularised model, read every input as of the scan day, and rebuilt the report on the audit record.

2.2. Data, outcome and analysis plans

MU-Glioma-Post (Mahmoud et al. 2025) was the development cohort. It is a collection from a single centre of MRI after surgery, with four-compartment masks (enhancing tumour, ET; non-enhancing core; surrounding FLAIR signal; resection cavity) and a clinical workbook. The workbook gives the day of each scan, surgery, RT, chemotherapy, first progression and death relative to diagnosis, and age, sex, grade, IDH and MGMT. LUMIERE (Suter et al. 2022) was the external cohort: glioblastoma follow-up scans with an expert RANO rating, automatic segmentations, IDH and MGMT, but no treatment dates. CFB-GBM (Moreau et al. 2025) supported a diffusion analysis: 264 patients imaged at RT planning and twice afterwards, with ADC, volumetric RANO 2.0 labels and released radiomics features, but no IDH or MGMT. We used only its released tables.

An index scan is an MRI before first documented progression and before later lines of treatment, with all four sequences, a mask and a known date. Each patient contributes every eligible scan, and every analysis keeps a patient's scans together. Grade 4 glioma was the primary population, fixed in advance (Table 1; study design in Figure 2; cohort flow in Supplementary S2). The outcome is documented first progression within 120 days of the scan. A scan counts as negative only if follow-up without progression reaches 120 days; otherwise it is censored, and death without progression censors at the day of death. Discrimination is the AUC at 120 days weighted by the inverse probability of censoring (IPCW AUC; Uno et al. 2007; Blanche et al. 2013). PRISM v1 counted censored scans as negatives; we use this naive label only to reproduce its evaluation. In LUMIERE, progression is the first expert rating of progressive disease.

Reporting follows TRIPOD+AI (Collins et al. 2024) and CLEAR (Kocak et al. 2023 [CHECK]; Supplementary S17). For each analysis we call prespecified (Table 4), the plan and its decision rule were written into the analysis document and code before any of its model outputs were seen; for the later analyses a dated copy of the plan was saved before the run (Supplementary S1). None of these plans was registered externally. Everything else is secondary or exploratory.

2.3. Predictors

All predictors are defined as of the scan day (Supplementary S4). Clinical: age, sex, IDH and MGMT, each with an unknown category. Timing: days since diagnosis, days since RT ended, a flag for the 12 weeks after RT and its interaction with MGMT methylation, and visit number. Treatment: whether RT and chemotherapy had started and whether RT was complete, the prescription, and the days of surgery and RT start. A treatment fact enters only once its documented start is on or before the scan day. Together these 23 predictors (31 model columns) form the *core model*. A compact alternative replaces timing and treatment with four phases of care: before RT, during RT, within 12 weeks after RT, and later. Imaging measurements are compartment volumes, morphology and change from earlier scans only. Radiomics are PRISM v1's T1c and FLAIR features and a corrected set covering all four sequences (Supplementary S5). Deep features are pooled from frozen SwinUNETR encoders with BraTS 2021 or BrainSegFounder weights (Cox et al. 2024 [CHECK]), with untrained weights as a control (Supplementary S6). Unknown values are imputed within each training fold with missingness indicators, and radiomics and deep features are clipped to the training fold's 1st and 99th percentiles.

2.4. Three ways to evaluate the same model

We evaluated each model in three ways, from the least to the most reliable.

The initial split. Our initial evaluation split came from PRISM v1's own split search: for a given seed, 20,000 random draws of 30 test patients, keeping the draw closest to 96 test scans and 53 positives. Seed 62 gave the split, and we reproduced the shipped model's stored test predictions to 1.1×10^{-15}. With the final cohort the split holds 126 training and 29 test patients (82 test scans). On it we fitted, once and without tuning, the core model (ridge and gradient boosting), the core model plus legacy radiomics, and age, sex, IDH and MGMT alone. Intervals come from 1,000 bootstrap draws over test patients.

500 splits. We reran v1's split search for seeds 0 to 499 and refitted four models on each split: PRISM's initial radiomics pipeline with its feature ranking redone (the honest estimate), the same pipeline with its 48 shipped features (optimistic, since those were chosen partly on other splits' test patients), and the core model with and without radiomics. We report where our initial split ranks and how the AUC of each split relates to the mix of phases in its test set (Supplementary S8).

Repeated cross-validation. All main estimates use one shared set of 5×5 repeated folds, grouped by patient and stratified by outcome and grade. Every fitted step (imputation, clipping, scaling, feature selection, stacking and hyperparameters) is fitted inside the training fold (Varma and Simon 2006; Cawley and Talbot 2010). The estimate is the mean over repeats of the pooled held-out IPCW AUC. 95% CIs and paired differences come from 1,000 bootstrap draws over patients, with all models scored on the same draws.

The treatment-date fix. Our earlier runs read treatment fields as of the end of follow-up, so a scan taken before RT carried the dose, start date and RT status of a course that had not yet begun (RT status was wrong in 23 scans; Supplementary S4). We report results before and after the fix (Table 2).

2.5. Ablation and phase of care

Each branch is added to, or removed from, the core model. The primary learner is ridge logistic regression with its penalty chosen by inner log loss; gradient boosting is secondary. Radiomics enters through 70 configurations (feature set, stability filter, selection, and either concatenation or a stacked score) and deep features through 32, with 16 untrained controls. The honest estimate picks one configuration inside each outer training fold by inner cross-validation (nested selection). With 70 to 80 uncorrected intervals, about two configurations are expected to exclude 0 in each direction by chance. The prespecified primary development analysis compared the core model with the core model plus radiomics (nested) in grade 4.

To test the phase of care, we built a ladder of bases without imaging (R0 clinical; R1 plus timing; R2 clinical plus phase; R3 the core model) and added four imaging blocks to each (I1 volumes; I2 all imaging measurements other than radiomics; I3 radiomics; I4 deep features). Models were trained and evaluated within the full cohort and within the surveillance cohort, the scans taken after RT was complete (grade 4: 167 scans, 84 patients, 83 events). The prespecified primary contrast was R2 + I2 against R2 in grade 4 surveillance scans, with a 100-draw permutation test that moves the imaging block between patients with the same number of scans. This plan was written into the code before the full run but not committed separately (Supplementary S11).

Censoring turned out to be concentrated before RT (Section 3.4), so we tested whether the phase signal survives it, with a plan committed to the public repository before any model was fitted (Supplementary S20). We split the phase before RT into RT pending and RT never documented, estimated the censoring weights separately within each of the five phases, and refitted after excluding patients whose RT was never documented (rule in Table 4).

2.6. Giving imaging every chance

When imaging added nothing, we tested five explanations. Except for the external cohort, each had a rule fixed before its own run (Table 4).

The label. Progressions documented on clinical grounds alone were censored, giving an imaging-defined label.

The segmentation. We checked volumes, labels and dates against their sources, then resegmented all 594 complete MU scans with the MONAI BraTS 2021 SwinUNETR and with BrainSegFounder. We then tested whether the new masks detect the progression scan itself better, and whether ET is larger at progression than at the previous scan.

The cohort. Models trained on all MU scans were applied unchanged to LUMIERE. Without treatment dates the transportable model was clinical variables, timing (days since surgery and a flag for the 3 months after RT) and volumes, trained on grade 4 (Supplementary S13).

The modality. In CFB-GBM, with one row per patient, the outcome was volumetric RANO 2.0 progression between the two scans after RT. Nested feature sets were clinical; plus tumour volume and its change; plus conventional T1-Gd and FLAIR radiomics; plus ADC radiomics, evaluated by 20 repeats of 5-fold cross-validation. ADC was tested with a 1,000-draw permutation test. We noted in advance that the volume block is partly circular, because the label is defined by the same volume (Supplementary S18).

The encoder. Frozen features may miss what a network trained on this label would find, so we fine-tuned the BraTS 2021 SwinUNETR encoder on the 120-day label inside the shared folds, with

a plan committed before training (Supplementary S22). The two deepest stages and a small head were trained on a cube around the lesion and cavity, with fixed hyperparameters and early stopping on patients held out from training. Within each outer training fold, four networks gave out-of-fold scores for stacking onto the core model and an averaged score for the test fold, 100 fine-tunes per arm. A copy with outcomes shuffled within each training fold showed what an uninformative score does to the stack. BrainSegFounder weights, all gliomas and a network that also sees the previous scan were secondary arms.

2.7. The audit layer

The audit layer wraps the deployed core model and adds no predictor (plan saved before the run; Supplementary S15). It locks the model with a model card (code version, hashes of every input table, each input's definition, source and as-of rule, training counts, intended and excluded uses) and writes one record per scan. The record holds each input with its source and the day it became known; the phase of care; the risk with a 95% interval from 200 bootstrap refits; the exact additive contribution of each input group to the log odds; a split conformal prediction set at $\alpha = 0.2$; and referral flags with reasons. A scan is flagged if it falls in the 12 weeks after RT, if an input lies outside the training range or a molecular marker is unknown, or if its conformal set is $\{0, 1\}$. The layer also tests selective prediction (does withholding the least confident scans help?), probes shortcuts by refitting without the indicators for unknown markers and with treatment or timing alone, and measures how stable the explanations are. Every fitted step, including the conformal calibration split (70/30 of training patients), stays inside the shared training folds.

Two quantities were prespecified (Table 4). **P1** is conformal coverage on test scans with a known outcome. **P2** is the AUC on scans kept at about 70% coverage, minus the AUC after removing the same number of scans at random within each phase.

2.8. The report generator

Each report is written from one scan's audit record and nothing else. Sections for scan context, inputs, risk (interval, a descriptive tier and the conformal set), contributions, referral flags and volume change are rendered directly from the record. A summary is written by a local language model (Qwen3.8-27B at a pinned revision, greedy decoding, run offline on our own GPUs) that sees only the record's fields. Every number must be copied from a field, every sentence must cite the fields it uses and every active flag must be stated; diagnoses, RANO categories, management advice and unhedged certainty are forbidden. From v1 we kept the change table, the explanation section and the tiered confidence language. We dropped the fixed alert threshold (set by hand on the initial split), the RANO rule engine and the printed outcome label (Supplementary S19).

A deterministic checker scores every sentence. Cited fields must exist, and every number must match a cited value. Every active flag must be mentioned, no inactive flag may be described as raised, the tier must match, and no forbidden pattern may appear. Prompts and checker were developed on the 59 scans of other grades and frozen before the evaluation on all 249 grade 4 scans. The prespecified primary quantity was the share of reports with no unsupported claim and every active flag mentioned (Wilson 95% CI; target in Table 4). We also ran v1's prompt on the same model and facts (arm B) and the grounded prompt on a smaller model (Qwen3.5-9B; arm C), and perturbed 30 records to check that the reports follow a changed risk or flag.

Faithful is not the same as useful, so we then wrote a second version of the summary (v2) with a plan committed before evaluation (Supplementary S21). No public set of post-treatment surveillance reports exists, so we used two public collections for form only: 230 glioma MRI reports from RadGenome-Brain MRI and the MRI descriptions of 181 published glioblastoma case reports (MultiCaRe). Like them, v2 has findings followed by a short impression. It adds the facts a reader needs next to the probability: how the risk compares with the model's average, the risk at the previous scan, why the phase of care matters, what the model did not see, and any input not known on the scan day. Its risk sentence cites a single field that holds the whole statement, which removes the citation failure of v1. Usefulness was defined in advance as eight items (risk with interval and horizon, reference frame, phase, drivers, what the model did not see, flags with reasons, change since the prior scan, look-ahead) and two safety checks (false reassurance, wording that disagrees with the number). v2 was to replace v1 only if it met four rules (Table 4) on 249 grade 4 records from a cross-validation repeat not used for any tuning. A model from a different family (Gemma 4 31B), blinded to version, rated the reports as an exploratory check; it is not a substitute for clinicians.

3. Results

3.1. Cohorts

Table 1 describes the cohorts. In the primary population, 249 grade 4 scans from 126 patients, 93 scans were followed by documented progression within 120 days. The scans spread across the phases of care: 76 were taken before RT, 6 during RT, 98 within 12 weeks after RT and 69 later.

3.2. The initial split scored high for every model

Rescored with the final cohort and labels, our initial evaluation split was generous to every model (Figure 3A; Table 2). PRISM's initial radiomics model reached 0.80, but the core model, which uses no imaging, reached 0.87, and age, sex, IDH and MGMT alone reached 0.79. The intervals were wide and overlapped.

One model went higher still. Before the treatment-date fix, gradient boosting on the core inputs scored 0.90 on this split, yet under repeated cross-validation the same model averaged 0.72 in grade 4 (Table 2, history rows). Two things had lifted it: treatment dates from the future, and the split. After the fix it still scored 0.87 on the split. A model that is really about 0.70 had looked like 0.90.

3.3. Why the split was easy

The test set rewarded any model that recognises the phase of care. Of its 60 negative scans, 36 were taken before RT, and none of those 36 was followed by documented progression, whereas 10 of the 68 scans before RT with a known outcome in the whole cohort were (15%; Figure 3B). So 60% of the positive–negative pairs set a progression scan against a fresh postoperative scan before RT. The core model ranked these pairs at 0.962 and the remaining pairs at 0.735. A rule with no training at all, "the scan was taken after RT", scored 0.740 on this test set.

Across 500 splits of the same design (Figure 4A; Table 2), PRISM's initial radiomics pipeline, reselecting its features on each split, scored higher on our initial split than on 92% of the others, and only 7% of splits reached its original 0.802. Refitting the 48 shipped features gave 0.815 on average,

an upper bound inflated by selection on other splits' test patients. Our split ranked the core model at the 91st percentile. The AUC of each split followed the mix of phases in its test set: for the core model, its Spearman correlation with the AUC of the "after RT" rule was +0.57 (Figure 4B). From composition alone, a regression predicted 0.851 for the core model on our split (observed 0.871, mean over splits 0.801). About two thirds of the split's excess came from its mix of scans, and the rest from its particular patients.

3.4. What carries the signal

Under repeated cross-validation the picture was consistent and modest. In grade 4, clinical features alone reached 0.585, adding timing 0.644, and adding treatment status 0.698: the core model (Figure 5A; Table 2). In all gliomas the same steps ended at 0.720. Gradient boosting did not beat ridge, and the core model was calibrated in the large (ratio of observed to expected events 0.96; slope 0.89; Supplementary S12).

Most of this signal is the phase of care. Four phase categories recovered most of the gain over clinical features (R2 0.664 against R0 0.585, a difference of +0.080, +0.005 to +0.152; Figure 6A), and treatment features alone scored 0.701, as high as the full core model. Within the surveillance cohort, where the phase varies little, no base model discriminated well (0.549 to 0.604), and the phase itself added nothing (−0.003).

The phase signal survives censoring, but it lies between phases. Of the 39 censored grade 4 scans, 38 were taken before RT, all from 37 patients whose RT and chemotherapy were never documented. These patients were older, were scanned around surgery and had a median follow-up of 36 days after the scan (317 days for patients with RT documented); 14 died without documented progression. Censoring weights estimated within each phase raised the gain of R2 over R0 from +0.080 to +0.124, and excluding these patients left +0.057, so the prespecified rule for a robust finding was met (Table 4). The single censoring curve was therefore conservative (the core model scores 0.749 with phase-specific weights). Two qualifications remain. If every censored scan had progressed, the gain would vanish (−0.044), which seems unlikely but cannot be excluded. And the phase separates scans between phases, not within them: counting only pairs within the same phase, R2 scored 0.577 against 0.595 for R0 (Supplementary S20).

No imaging branch showed a measurable gain (Table 3; Figure 5B). In the prespecified primary analysis, nested radiomics changed the grade 4 core model by −0.032, and none of the 70 configurations had a CI above 0. Frozen pretrained encoders changed it by −0.004. Their features did carry an image signal (0.652 alone, against 0.566 with untrained weights), but it tracked tumour volume ($r = 0.58$) and added nothing to the core model. Radiomics also made the model overconfident (calibration slope 0.60) and lowered its net benefit at a 30% threshold by 0.020.

A model can give imaging credit without needing it. In models that included volumes, morphology and change, these groups took about half of the SHAP credit (45% with ridge, 51% with boosting), yet the models scored no better than the core model with the same learner (ridge 0.692 against 0.698; boosting 0.694 against 0.687; Figure 5C).

3.5. Imaging, given every chance

Within the surveillance cohort, where the phase cannot do the work, the prespecified primary contrast added all imaging measurements to clinical and phase variables. The AUC moved from

0.549 to 0.610 (permutation p = 0.21; Figure 6B; Table 4). The permutation null itself averaged +0.028, because adding 44 columns of noise to a weak base raises its AUC. Radiomics gave the same +0.061 (−0.041 to +0.157). The hypothesis that imaging adds value after RT was not supported.

The label was not the bottleneck. Censoring the 12 of 93 grade 4 events that were documented on clinical grounds alone gave the core model plus imaging 0.722 against 0.696. The rule was not met.

Nor was the segmentation. Our volumes reproduced the dataset's own table ($r \geq 0.996$ for every compartment), and labels and dates were aligned. Yet imaging separated the progression scan itself from verified negatives with an AUC of only 0.68, and ET volume was larger at the progression scan than at the scan before in only 49% of 74 pairs. Resegmentation with the BraTS 2021 model agreed with the original masks (median Dice 0.80 for ET) and reduced the noise between closely spaced scans, but detection fell from 0.681 to 0.665, and ET was larger at progression in 45% of pairs. The rule was not met (Supplementary S14).

The external cohort showed no imaging effect either. On LUMIERE the locked model with volumes scored 0.608 (0.469–0.739) with ridge and 0.620 with boosting, against 0.667 and 0.675 in MU, while clinical variables and timing alone scored 0.610 and 0.632. PRISM's initial radiomics model scored 0.613. The time to the next scan alone scored 0.788, higher than any locked model, which shows how much this label depends on scheduling (Supplementary S13).

Diffusion did not change the result. In 77 CFB-GBM patients with ADC (19 events), ADC radiomics added +0.007 to conventional radiomics; the rule was not met. Tumour volume and its change added 0.15 to 0.19, but that block is partly circular with the label, and conventional texture added nothing beyond volume.

Nor did training the encoder on the task. The fine-tuned image score alone reached 0.644 (0.559–0.727) in grade 4, close to the frozen features' 0.652, and it again tracked tumour volume ($r = 0.57$) more than the core model ($r = 0.13$). Added to the core model it gave 0.708, a difference of +0.009; the rule was not met (Tables 3 and 4). The shuffled-outcome control scored 0.476 alone, as intended, but its stack gained +0.005, slightly above the limit of 0 that we had set for a valid run. An uninformative score therefore buys about half of the observed gain, which weakens any reading of the +0.009 as signal. BrainSegFounder weights gave −0.005, all gliomas +0.017 and the network that also saw the previous scan +0.014. Fine-tuning tightened the interval: gains of about 0.045 or more can now be excluded in grade 4, against 0.09 for frozen features (Supplementary S22).

3.6. Auditing a modest model

The audit layer reproduced the core model's predictions exactly (largest difference 1.7×10^{-16}), and the contributions summed to every logit. **P1 was met** (Table 4). Coverage failed where it mattered most, however. Almost every scan before RT received the set {0}, so all 5 grade 4 progressions before RT fell outside their sets.

P2 was not met. Withholding the least confident 30% of scans gave the remaining scans an AUC of 0.731, against 0.698 after random removal, but removing the same number of scans at random within each phase gave 0.725 (Figure 7B; Table 4). Abstention helped only by changing the phase mix: it kept 97.6% of scans taken before RT. Referral flags applied to 72.6% of grade 4 scans.

The audit's value was what it found. First, a missingness shortcut: IDH was unknown in none of 93 progression scans but in 6 of 117 scans without progression, and removing the indicators

for unknown markers lowered the grade 4 AUC from 0.698 to 0.676 (difference 0.023, 0.001 to 0.047). Second, the core model is a treatment-course model: treatment features alone scored 0.701. Third, a look-ahead: IDH and MGMT were used at 8 grade 4 scans taken 1 to 58 days before the diagnosis date. Fourth, the same shortcut explained most of the miscalibration on LUMIERE, where IDH was unknown in 40.9% of scans. Without the indicators for unknown markers, the ratio of observed to expected events there moved from 1.33 to 0.89. Fifth, the censoring analysis found one more leak. The flag for the 12 weeks after RT is missing when the RT end date is missing, and the missingness indicator added during imputation told a scan whose RT would be documented later apart from one whose RT never would be. None of these problems showed in the cross-validated AUC.

The corrected model. Removing the indicators for unknown markers, masking IDH and MGMT before the diagnosis date and coding the early window as known on the scan day gave the corrected core model: 0.678 (0.596–0.755) in grade 4, −0.020 (−0.044 to 0.000) against the original, with a ratio of observed to expected events of 1.04 and a calibration slope of 0.85 (0.706 in all gliomas). Almost all of the drop came from the unknown-marker indicators. On LUMIERE, the same corrections left discrimination unchanged within its wide interval (0.635 against 0.608) but fixed the calibration: observed to expected 0.99 (was 1.33), slope 0.97 (was 0.38). We deploy the corrected model. Restricted to patients on a documented RT pathway, where nearly all censoring disappears, the original and corrected core models scored 0.654 and 0.625.

3.7. Reports that can be checked

Of 249 grade 4 grounded reports, 236 had no unsupported claim and mentioned every active flag, one report short of the 0.95 target (Table 4). All 13 failures had the same cause: the risk sentence stated the 120-day horizon without citing the horizon field, although the number was correct. No grounded report omitted any of the 269 active flags, raised an inactive one or contained forbidden content (Figure 7A).

The prompt mattered more than the model. Given the same facts, v1's prompt passed 171 of 249 reports against its own input (0.687), and every one of its reports contained forbidden content: management advice in 99%, a diagnosis in 51% and a RANO category in 41%. It omitted 48 of the 269 active flags. The grounded prompt on the smaller model passed 243 of 249 (0.976), with one real violation, a hedged diagnosis. The grounded reports followed 27 of 30 perturbed records. An AI assistant read 15 grounded reports against their records and found no factual error; no clinician has read them yet (Supplementary S19).

Version 2 is more useful and fully faithful. On 249 fresh records, every v2 report passed the checker and 247 met every usefulness item, against none for v1 (Table 5; Figure 7A). The gain came from content that v1 lacked: v1 never said what the model did not see or that an input was not yet known on the scan day, and it named the main drivers in only 32 reports. Every prespecified rule was met, and v2 is now PRISM's default. The cost is length, about a minute to read against 40 seconds, although v2 reads at a lower grade level. The blinded judge preferred v2 in both presentation orders for 152 scans and v1 for 5, but it gave the top score on most criteria to both versions and missed omissions, so we do not rely on it (Supplementary S21).

4. Discussion

We built PRISM to predict progression from surveillance MRI, and its evaluation changed what PRISM is. Three lessons emerged.

A single split can measure the split. Our initial 0.80 was real for that test set, but the test set was easy: 60% of its pairs set a progression scan against a fresh postoperative scan before RT, a contrast that any model able to recognise the phase of care will win. The same split lifted the core model, which scores 0.698 under repeated cross-validation, to 0.871, and lifted a boosting model with leaked treatment dates from 0.723 to 0.901. This adds a mechanism to the known instability of single splits (An et al. 2021). When phases with very different event rates are mixed, the composition of a small test set explains much of the variance (about two thirds of this split's excess), and a split search that matches the number of scans and positives does not control it. We suggest four reporting rules for this task: repeated grouped cross-validation; the distribution over splits whenever a single split is reported; results within phases of care; and a clinical comparator that knows the treatment course.

The phase of care is a confounder. A scan before RT is rarely followed by documented progression within 120 days. A scan in the 12 weeks after RT often is, partly because of pseudoprogression and the follow-up it triggers (Brandes et al. 2008; Taal et al. 2008; Wen et al. 2023). Four phase categories recovered most of the core model's gain, and treatment status alone matched the whole model. Two published clinical comparators also found time since RT influential (Gomaa et al. 2025; Leone et al. 2023). Imaging models that pool scans from all phases can therefore report high AUCs for recognising the phase rather than for tumour biology. This is why PRISM places every scan in its phase and flags the 12 weeks after RT. The phase signal needs care, however. Censoring was concentrated in patients whose RT was never documented; the signal survived weights that account for this, but it ranks scans between phases rather than within them, and part of it reflects who stays in follow-up. PRISM's intended use is therefore restricted to patients on a documented RT pathway, where its AUC is about 0.65 and within-phase performance should be reported.

Imaging, as tested here, showed no gain. We found no pipeline defect, yet even at the scan where progression was documented, volumes separated it from negatives with an AUC of only 0.68. Three features of the data explain this. The label is a documentation event: one in eight grade 4 events was documented on clinical grounds alone, and the 120-day label mostly means that the next scan was the progression scan. Volume series were noisy, with ET changing as much over days as over months. And the cohort is small: by the Riley criteria even the 31-column core model would need about ten times more scans (Riley et al. 2020). Cleaning the label and resegmenting moved imaging up by about 0.03, and imaging added 0.06 within the surveillance cohort, and the upper confidence bounds reach 0.02 to 0.16. Effects of that size cannot be excluded; our data cannot detect them. Training the encoder on the label narrowed this: its gain was +0.009, and gains above about 0.045 are now unlikely for deep features in grade 4. Diffusion did not change this in CFB-GBM, although that test was small. Where imaging performs well in the literature, it uses diffusion and perfusion to classify a lesion that is already suspicious (Kim et al. 2019; Taylor et al. 2022), a different question from forward risk at every scan. We keep PRISM's imaging branches as optional modules so that larger cohorts can test them again. Attribution is no guide here: imaging took half of the SHAP credit without improving discrimination, so a claim of imaging value needs a paired difference with an interval.

What PRISM offers. At an AUC of about 0.70, PRISM does not offer better triage. Abstention and referral raised the AUC of the remaining scans only by changing their phase mix, and the conformal sets met their target mainly by calling every scan before RT negative. What PRISM does offer is a record of what the model knew on the scan day, where the scan sits in the treatment course, how uncertain the estimate is and which inputs drove it. Running the audit found a missingness shortcut, a molecular look-ahead and the cause of the external miscalibration, none of which the AUC revealed. Grounding the language model in that record, and checking each report against it, turned a report that issued RANO verdicts and management advice into one that restates the record, and the second version adds what a reader needs next to the number: how the risk compares with the average, how it changed since the last scan, what the model could not see and why the phase matters. Whether clinicians find it useful is the next question; we provide a protocol for a randomised reader study (Supplementary S21). We think this is the honest use of a modest model: to make its reasoning and its failure modes visible, not to claim better decisions.

Comparison with other work. The grade 4 core model (0.698) is close to clinical comparators for neighbouring tasks (0.69 in Leone et al. 2023; 0.727 externally in Gomaa et al. 2025) and to the nested result for a volume label on the same dataset (0.728; Rai 2026 [CHECK]). It is below the single test set result of 0.80 on MU-Glioma-Post (Christodoulou et al. 2026), which used a different label (the progression status of each scan) and which we did not re-evaluate. Higher AUCs for forward recurrence prediction elsewhere use different cohorts, horizons and units, and some fall externally (0.887 to 0.762 in Chen et al. 2026).

Limitations

The development cohort comes from one centre, with 126 grade 4 patients and 93 events at 120 days, far below recommended sample sizes. The outcome is documented progression from routine records, not a blinded RANO read, and its timing depends on scheduling. MU images carry no acquisition metadata, so scanner effects could not be studied. Censoring was concentrated in patients whose RT was never documented; phase-specific censoring weights and their exclusion left the phase finding intact, but censoring that depends on the outcome itself cannot be ruled out. Our bootstrap intervals resample fixed held-out predictions and do not include the variability of refitting, so they are likely too narrow. Death was treated as censoring rather than as a competing risk. The external cohort is small, uses a different segmenter and lacks treatment dates, so the full core model was not tested externally. The label and segmentation tests were added after the imaging result was known, although each had its rule fixed before its own run, and the plan for the phase ladder was not committed separately. Radiomics used a fixed bin count rather than the fixed bin width recommended by IBSI (Zwanenburg et al. 2020). The fine-tuned encoder was trained on about 100 scans per network with fixed hyperparameters and stopped early, and its permuted-label control slightly missed its validity limit. The ablation and phase analyses use the original core model; the corrected model, which removes the leaks found by the audit, scores 0.020 lower. The faithfulness checker is lexical: it verifies numbers, citations, flags and forbidden patterns but not meaning, which was checked only by an AI assistant reading 15 reports. The reports come from one cohort, the public reference reports describe pre-operative or published cases rather than routine surveillance, our usefulness criteria were written with v2 in mind, and no clinician has assessed the reports.

Conclusion

PRISM's initial AUC of 0.80 came from one favourable split. Across splits its radiomics pipeline averaged 0.72, and its core of clinical and treatment-course variables reached about 0.70 under repeated cross-validation (0.68 after removing leaks found by its own audit). MRI radiomics, deep features (frozen or fine-tuned), resegmentation, a cleaner label and diffusion added no measurable discrimination. The phase of care drives discrimination on this task and should be reported as a stratum and treated as a confounder. PRISM is deployed as its corrected core model for patients on a documented RT pathway, and every prediction is recorded, explained and reported in a form that can be checked.

Acknowledgements

[PENDING AUTHOR DECISION].

Disclosure statement

[PENDING AUTHOR DECISION: competing interests, or "The authors report there are no competing interests to declare."] If any author serves on the editorial or advisory board of this journal, state it here.

Declaration of generative AI use

[DRAFT, PENDING AUTHOR CONFIRMATION:] Generative AI is part of the system studied. PRISM's report generator uses a local language model (Qwen3.8-27B; Qwen3.5-9B in a comparison arm), and a second local model (Gemma 4 31B) rated reports in an exploratory check; all ran offline on the authors' hardware, saw only derived audit records, and are described in Sections 2.8 and 3.7. An AI assistant (Claude, Anthropic) was used to write and run analysis code, to read 15 generated reports against their records (Supplementary S19), and to draft and edit the manuscript under the authors' direction. Patient data did not leave the authors' computing cluster. The authors reviewed all code, results and text and take full responsibility for the content of this article.

Funding

[PENDING AUTHOR DECISION: "This work was supported by the [Funding Agency] under Grant [number]." If there was no funding: "No funding was obtained for this work."]

Ethics statement

[PENDING AUTHOR DECISION: secondary analysis of public de-identified datasets (MU-Glioma-Post, LUMIERE, CFB-GBM); state the institutional determination or waiver and cite the source datasets' approvals.] No registered protocol. Analysis plans and dated plan snapshots are in the repository (Supplementary S1). Patient and public involvement: none.

Data availability statement

[PENDING AUTHOR DECISION: MU-Glioma-Post (TCIA, doi:10.7937/7K9K-3C83), LUMIERE (figshare; Suter et al. 2022) and CFB-GBM (TCIA, doi:10.7937/v9pn-2f72) are public; derived tables, fold assignments and locked model files; state what is released and give a DOI or other persistent identifier.] Code: [PENDING: public repository link and release tag.]

Supplemental material

Supplemental data for this article (Supplementary S1–S22, Tables S1–S43) are provided as a separate file.

PRISM versions

PRISM v1 and its initial evaluation on a single split were described in an unpublished internal report [PENDING AUTHOR DECISION: by whom; whether to cite it]; this manuscript supersedes it. “Our initial evaluation split” and “PRISM’s initial radiomics model” refer to the split and the radiomics model of that report.

Notes on contributors

[PENDING: a short biographical note for each author (no more than 200 words each).]

References

Abler D, Pusterla O, Joye-Kühnis A et al. 2025. The added value for MRI radiomics and deep-learning for glioblastoma prognostication compared to clinical and molecular information. [Preprint, arXiv]. `https://doi.org/10.48550/arXiv.2507.15548`

An C et al. 2021. Radiomics machine learning study with a small sample size: Single random training-test set split may lead to unreliable results. PLOS ONE. 16(8):e0256152. https://doi.org/10.1371/journal.pone.0256152

Blanche P, Dartigues JF, Jacqmin-Gadda H. 2013. Estimating and comparing time-dependent areas under receiver operating characteristic curves for censored event times with competing risks. Stat Med. 32(30):5381–5397. https://doi.org/10.1002/sim.5958

Booth TC, Grzeda M, Chelliah A et al. 2022. Imaging biomarkers of glioblastoma treatment response: A systematic review and meta-analysis of recent machine learning studies. Front Oncol. 12:799662. https://doi.org/10.3389/fonc.2022.799662

Brandes AA, Franceschi E, Tosoni A et al. 2008. MGMT promoter methylation status can predict the incidence and outcome of pseudoprogression after concomitant radiochemotherapy in newly diagnosed glioblastoma patients. J Clin Oncol. 26(13):2192–2197. https://doi.org/10.1200/JCO.2007.14.8163

Cawley GC, Talbot NLC. 2010. On over-fitting in model selection and subsequent selection bias in performance evaluation. J Mach Learn Res. 11(70):2079–2107. `https://jmlr.org/papers/v11/cawley10a.html`

Chen C, Guo F, Huang D et al. 2026. Predicting recurrence risk of glioblastoma based on preoperative-postoperative longitudinal MRI: A multicenter study. Bioengineering. 13(6):668. https://doi.org/10.3390/bioengineering13060668

Christodoulou RC, Vamvouras G, Pitsillos R, Solomou EE, Georgiou MF. 2026. Explainable radiomics with probability calibration for postoperative glioblastoma surveillance. Eur J Radiol Artif Intell. 5:100074. https://doi.org/10.1016/j.ejrai.2026.100074

Collins GS, Moons KGM, Dhiman P et al. 2024. TRIPOD+AI statement: Updated guidance for reporting clinical prediction models that use regression or machine learning methods. BMJ. 385:e078378. https://doi.org/10.1136/bmj-2023-078378

Cox J et al. 2024. BrainSegFounder [full title to be completed]. Med Image Anal. CHECK. https://doi.org/10.1016/j.media.2024.103301

Demircioğlu A. 2021. Measuring the bias of incorrect application of feature selection when using cross-validation in radiomics. Insights into Imaging. 12(1):172. https://doi.org/10.1186/s13244-021-01115-1

Gomaa A, Huang Y, Stephan P et al. 2025. A self-supervised multimodal deep learning approach to differentiate post-radiotherapy progression from pseudoprogression in glioblastoma. Sci Reports. 15:17133. https://doi.org/10.1038/s41598-025-02026-7

Kapoor S, Narayanan A. 2023. Leakage and the reproducibility crisis in machine-learning-based science. Patterns. 4(9):100804. https://doi.org/10.1016/j.patter.2023.100804

Kim JY, Park JE, Jo Y et al. 2019. Incorporating diffusion- and perfusion-weighted MRI into a radiomics model improves diagnostic performance for pseudoprogression in glioblastoma patients. Neuro-Oncology. 21(3):404–414. https://doi.org/10.1093/neuonc/noy133

Kocak B, Baessler B, Bakas S et al. 2023. CheckList for EvaluAtion of Radiomics research (CLEAR): A step-by-step reporting guideline for authors and reviewers endorsed by ESR and EuSoMII. Insights into Imaging. 14:75. CHECK. https://doi.org/10.1186/s13244-023-01415-8

Leone R, Meredig H, Foltyn-Dumitru M et al. 2023. Assessing the added value of apparent diffusion coefficient, cerebral blood volume, and radiomic magnetic resonance features for differentiation of pseudoprogression versus true tumor progression in patients with glioblastoma. Neuro-Oncology Adv. 5(1):vdad016. https://doi.org/10.1093/noajnl/vdad016

Mahmoud E, Gass J, Dhemesh Y et al. 2025. MU-Glioma Post: A comprehensive dataset of automated MR multi-sequence segmentation and clinical features. Sci Data. 12(1):1847. https://doi.org/10.1038/s41597-025-06011-7

Moreau NN et al. 2025. Pre and post treatment MRI and radiotherapy plans of patients with glioblastoma: the CFB-GBM cohort (CFB-GBM). [The cancer imaging archive, version 4]. Data set; metadata from DataCite. https://doi.org/10.7937/v9pn-2f72

Rai AK. 2026. Uncertainty gated residual learning with patient level bagging for longitudinal glioma progression prediction. J Intell Decis Mak Inf Sci. 3. CHECK. https://doi.org/10.59543/jidmis.v3.2236

Riley RD, Ensor J, Snell KIE et al. 2020. Calculating the sample size required for developing a clinical prediction model. BMJ. 368:m441. https://doi.org/10.1136/bmj.m441

Suter Y, Knecht U, Valenzuela W et al. 2022. The LUMIERE dataset: Longitudinal glioblastoma MRI with expert RANO evaluation. Sci Data. 9(1):768. https://doi.org/10.1038/s41597-022-01881-7

Taal W, Brandsma D, de Bruin HG et al. 2008. Incidence of early pseudo-progression in a cohort of malignant glioma patients treated with chemoirradiation with temozolomide. Cancer. 113(2):405–410. https://doi.org/10.1002/cncr.23562

Taylor C, Ekert JO, Sefcikova V, Fersht N, Samandouras G. 2022. Discriminators of pseudoprogression and true progression in high-grade gliomas: A systematic review and meta-analysis. Sci Reports. 12:13258. https://doi.org/10.1038/s41598-022-16726-x

Uno H, Cai T, Tian L, Wei LJ. 2007. Evaluating prediction rules for t-year survivors with censored regression models. J Am Stat Assoc. 102(478):527–537. https://doi.org/10.1198/016214507000000149

van Griethuysen JJM, Fedorov A, Parmar C et al. 2017. Computational radiomics system to decode the radiographic phenotype. Cancer Res. 77(21):e104–e107. https://doi.org/10.1158/0008-5472.CAN-17-0339

Varma S, Simon R. 2006. Bias in error estimation when using cross-validation for model selection. BMC Bioinform. 7:91. https://doi.org/10.1186/1471-2105-7-91

Welch ML, McIntosh C, Haibe-Kains B et al. 2019. Vulnerabilities of radiomic signature development: The need for safeguards. Radiother Oncol. 130:2–9. https://doi.org/10.1016/j.radonc.2018.10.027

Wen PY, Macdonald DR, Reardon DA et al. 2010. Updated response assessment criteria for high-grade gliomas: Response Assessment in Neuro-Oncology Working Group. J Clin Oncol. 28(11):1963–1972. https://doi.org/10.1200/JCO.2009.26.3541

Wen PY, van den Bent M, Youssef G et al. 2023. RANO 2.0: Update to the Response Assessment in Neuro-Oncology criteria for high- and low-grade gliomas in adults. J Clin Oncol. 41(33):5187–5199. https://doi.org/10.1200/JCO.23.01059

Yu AC, Mohajer B, Eng J. 2022. External validation of deep learning algorithms for radiologic diagnosis: A systematic review. Radiol Artif Intell. 4(3):e210064. https://doi.org/10.1148/ryai.210064

Yun YC, Jende JME, Garhöfer F et al. 2025. Combined peritumoral radiomics and clinical features predict 12-month progression free survival in glioblastoma. J Neuro-Oncology. 174:111–120. https://doi.org/10.1007/s11060-025-05037-6

Zwanenburg A, Vallières M, Abdalah MA et al. 2020. The Image Biomarker Standardization Initiative: Standardized quantitative radiomics for high-throughput image-based phenotyping. Radiology. 295(2):328–338. https://doi.org/10.1148/radiol.2020191145

Tables

Table 1. Cohort characteristics. Index scans are MRIs before first documented progression. Values are n (%) or median [IQR]. Patient rows use each patient's first index scan. Full table with definitions: Supplementary Table S2.

	MU all gliomas	MU grade 4	LUMIERE
Patients / index scans	155 / 308	126 / 249	61 / 132
Index scans per patient, median (range)	2 (1–6)	2 (1–6)	2 (1–10)
Age, years	61 [47–69]	62 [53–70]	62 [53–70]
Male sex	80 (52%)	66 (52%)	32 (52%)
IDH wild type / mutant / unknown	120 (77%) / 22 (14%) / 13 (8%)	112 (89%) / 8 (6%) / 6 (5%)	38 (62%) / 1 (2%) / 22 (36%)
MGMT methylated / unmethylated / unknown	53 (34%) / 73 (47%) / 29 (19%)	45 (36%) / 62 (49%) / 19 (15%)	26 (43%) / 28 (46%) / 7 (11%)
WHO grade 4	126 (81%)	126 (100%)	61 (100%)
Radiotherapy documented	102 (66%)	89 (71%)	not recorded
Initial chemotherapy documented	96 (62%)	88 (70%)	not recorded
Days from origin to scan[a]	97 [46–186]	96 [54–182]	119 [82–226]
RT completed by scan day (scans)	185 (60%)	167 (67%)	not recorded
Within 12 weeks after RT[b] (scans)	104 (34%); unknown 94[c]	98 (39%); unknown 65[c]	27 (20%)
Gap to next MRI, days	77 [50–103]	74 [49–98]	91 [77–117]
Death without documented progression (scans)	22 (7%)	20 (8%)	7 (5%)
120 d: progression / no progression / censored	100 / 156 / 52	93 / 117 / 39	47 / 76 / 9
90 d	65 / 195 / 48	62 / 152 / 35	24 / 101 / 7
180 d	144 / 105 / 59	131 / 75 / 43	62 / 61 / 9

[a] MU: days since diagnosis; LUMIERE: days since first surgery.
[b] MU: 0–84 days after the documented RT end date; LUMIERE: rater's "< 3 months after RT" flag.
[c] RT never documented (92 scans, all gliomas; 63, grade 4) or RT end date missing (2 scans). Every scan is still assigned a phase of care (Section 2.3).

Table 2. The same models on a single split, across 500 splits and under repeated cross-validation. IPCW AUC(120 d) with 95% bootstrap over patients CI unless stated. Single split = our initial evaluation split (all gliomas; 29 test patients, 82 scans). 500 splits = PRISM v1's split design (naive label; for PRISM's initial radiomics model, its pipeline with features reselected per split). Core model = PRISM's deployed model of clinical, timing and treatment variables (the reference for every imaging comparison). 5 × 5 CV = repeated cross-validation grouped by patient on the final cohort. History rows document the treatment-date fix on the earlier 30-patient / 84-scan version of the split; "before fix (leaky)" means that future RT dates and RT status entered the treatment and timing features.

Model	Single split: naive AUC	Single split: IPCW AUC [95% CI]	Δ vs core ridge [95% CI]	500 splits: naive mean ± SD (2.5–97.5%)	5 × 5 CV, all gliomas	5 × 5 CV, grade 4
PRISM initial radiomics model	0.802	0.817 [0.690, 0.920]	−0.034 [−0.186, +0.109]	0.720 ± 0.055 (0.610–0.819)	n/a (fixed model)	n/a (fixed model)
Core model (ridge)	0.871	0.851 [0.735, 0.947]	–	0.801 ± 0.052 (0.694–0.895)	0.720 [0.638, 0.792]	0.698 [0.615, 0.774]
Core model (GBM)	0.844	0.824 [0.690, 0.916]	−0.027 [−0.127, +0.062]	not run	0.718 [0.642, 0.792]	0.687 [0.601, 0.768]
Core + legacy radiomics	0.862	0.845 [0.746, 0.936]	−0.006 [−0.081, +0.090]	0.799 ± 0.053 (0.674–0.892)	0.743 [0.678, 0.803]	0.703 [0.627, 0.774]
Clinical only (ridge)	0.791	0.782 [0.638, 0.897]	−0.069 [−0.215, +0.058]	not run	0.619 [0.540, 0.696]	0.585 [0.496, 0.666]
History rows (30 patients / 84 scans; earlier cohort and inputs)						
PRISM initial radiomics model	0.802	0.811 [0.694, 0.919]	not computed	–	–	–
Core ridge, before fix (leaky)	0.870	0.844 (no CI)	not computed	not run	0.719 [0.648, 0.788][c]	0.694 [0.613, 0.775][c]
Core ridge, after the fix	0.880	0.860 [0.730, 0.943]	not computed	not run	0.749 [0.675, 0.820][d]	0.729 [0.648, 0.811][d]
Core GBM, before fix (leaky)	0.901	0.881 (no CI)	not computed	not run	0.743 [0.666, 0.816][c]	0.723 [0.633, 0.806][c]
Core GBM, after the fix	0.869	0.855 (no CI)	not computed	not run	0.740 [0.663, 0.815][d]	0.731 [0.641, 0.813][d]

[c] Inputs before the fix, cohort before the label fixes, earlier repeat-averaged estimator.
[d] Intermediate inputs after the treatment-date fix, before the label fixes; repeat-averaged estimator. History-row CV values are not comparable with the final rows.

Table 3. Do PRISM's imaging branches add to its core model? Paired ΔAUC with 95% bootstrap over patients CI. Reference: PRISM's core model (clinical + timing + treatment, ridge) unless stated in the section heading. Grade 4 (G4) is primary. * = CI excludes 0 (uncorrected; 3 of the 32 intervals in Figure 5B; the CFB-GBM row is added here). The decision rules and their answers are in Table 4.

Analysis	G4 ΔAUC [95% CI]	All gliomas ΔAUC [95% CI]
Radiomics branch vs core model		
Nested over 70 configurations (prespecified primary)	−0.032 [−0.084, +0.017]	+0.028 [−0.017, +0.071]
Legacy top-10, concatenated (single configuration)	+0.005 [−0.059, +0.067]	+0.024 [−0.026, +0.075]
Legacy PCA-10, stacked (best post hoc, optimistic)	+0.023 [−0.013, +0.057]	+0.035 [+0.002, +0.068]*
Frozen deep encoder branch vs core model		
Nested over 32 (BraTS21 SwinUNETR + BrainSegFounder)	−0.004 [−0.069, +0.063]	+0.011 [−0.032, +0.056]
BraTS21 SwinUNETR, lesion PCA-10 stacked	+0.016 [−0.029, +0.059]	+0.017 [−0.014, +0.050]
BrainSegFounder, lesion PCA-10 stacked	+0.014 [−0.029, +0.057]	+0.024 [−0.013, +0.064]
Untrained encoder, nested (control)	−0.049 [−0.085, −0.014]*	−0.009 [−0.032, +0.009]
Fine-tuned deep encoder branch vs core model		
BraTS21 SwinUNETR, fine-tuned (prespecified primary)	+0.009 [−0.022, +0.040]	+0.017 [−0.009, +0.044]
BrainSegFounder, fine-tuned	−0.005 [−0.030, +0.019]	–
BraTS21 SwinUNETR, fine-tuned with the previous scan	+0.014 [−0.010, +0.040]	–
BraTS21 SwinUNETR, shuffled outcomes (control)	+0.005 [−0.006, +0.015]	–
Imaging-defined label (progression on clinical grounds alone censored)		
Core + volumes/change vs core (prespecified rule)	+0.027 [−0.033, +0.082]	+0.008 [−0.036, +0.049]
Surveillance scans (after RT) only		
Clinical + phase + imaging vs clinical + phase (prespecified primary)	+0.061 [−0.038, +0.156]	+0.033 [−0.054, +0.128]
Core + imaging measurements vs core	+0.017 [−0.069, +0.093]	+0.005 [−0.075, +0.082]
Core + radiomics (nested) vs core	+0.005 [−0.070, +0.075]	+0.024 [−0.030, +0.076]
Resegmentation (reference: same model on MU masks)		
Clinical + phase + imaging, BraTS21 SwinUNETR masks	+0.030 [−0.008, +0.069]	+0.029 [−0.010, +0.069]
Clinical + phase + imaging, BrainSegFounder masks	+0.008 [−0.023, +0.040]	+0.003 [−0.025, +0.032]
LUMIERE, locked (reference: clinical + timing)		
+ volumes	−0.001 [−0.073, +0.057]	−0.006 [−0.075, +0.055]
+ deep features	+0.057 [−0.132, +0.235]	+0.064 [−0.011, +0.144]
+ volumes + deep vs + volumes	+0.089 [−0.026, +0.215]	+0.050 [+0.002, +0.109]*
CFB-GBM (77 patients, 19 events; reference: conventional radiomics)		
+ ADC radiomics (prespecified primary)	+0.007 [−0.019, +0.035] (permutation p = 0.151)	–

Table 4. Prespecified questions, rules and answers. Each rule was fixed before its analysis was run (Supplementary S1). Grade 4 glioma unless stated. ΔAUC is the paired difference in IPCW AUC at 120 days with 95% bootstrap CI over patients; "CI > 0" means the interval excludes 0. Core = PRISM's core model; R0 = clinical; R2 = clinical + phase of care; I2 = all imaging measurements except radiomics.

Question	Rule	Result	Answer
Does imaging add to PRISM's core model?			
Radiomics (primary development analysis)	Core + nested radiomics vs core; report the paired ΔAUC	−0.032 [−0.084, +0.017]	No gain
Imaging after RT, where the phase cannot help	R2 + I2 vs R2 in surveillance scans; permutation test	+0.061 [−0.038, +0.156]; p = 0.21	Not supported
Is the label the bottleneck?	Imaging-defined label: imaging gains ≥ 0.03 with CI > 0	+0.027 [−0.033, +0.082]	Not met
Is the segmentation the bottleneck?	New masks raise detection of the progression scan by ≥ 0.05 with CI > 0, or ET is larger at progression in > 65% of pairs	Detection 0.681 to 0.665; ET larger in 45%	Not met
Does diffusion add? (CFB-GBM)	ADC radiomics adds ≥ 0.03 to conventional radiomics with CI > 0	+0.007 [−0.019, +0.035]; p = 0.151	Not met
Does fine-tuning the encoder help?	Core + fine-tuned score gains ≥ 0.03 with CI > 0; validity: shuffled-outcome control gains ≤ 0	+0.009 [−0.022, +0.040]; control +0.005 [−0.006, +0.015]	Not met; control flagged
Is the phase of care an artefact of censoring?			
Phase signal under censoring	R2 − R0 ≥ 0.040 with phase-specific censoring weights and after excluding patients whose RT was never documented	+0.124 [+0.039, +0.232]; +0.057 [+0.009, +0.117]	Robust
Does the audit layer work?			
P1: conformal coverage	Coverage ≥ 0.78 on test scans with a known outcome	0.841 [0.790, 0.885]	Met
P2: selective prediction	Abstaining on the least confident 30% beats random removal within phase, CI > 0	+0.006 [−0.023, +0.036]	Not met
Are the reports faithful and useful?			
Report v1: faithfulness	≥ 0.95 of reports with no unsupported claim and every active flag stated	236/249 = 0.948 [0.913, 0.969]	Not met (one report short)
Report v2: replaces v1	Faithful in ≥ 95%; every usefulness item in ≥ 90%; paired usefulness gain with CI > 0; no safety failure	100%; 99.2%; +0.325 [+0.312, +0.337]; none	Met

Table 5. Report versions 1 and 2 on the same 249 grade 4 records. The same language model, records and deterministic checker; records from a cross-validation repeat not used for any tuning. Usefulness items were fixed before the evaluation. Differences: paired patient-cluster bootstrap (2,000 draws).

	Version 1	Version 2
Faithful (no unsupported claim, every active flag stated)	238 (95.6%)	249 (100%)
Every applicable usefulness item met	0	247 (99.2%)
Mean usefulness score (0 to 1)[a]	0.674	0.999
Usefulness items, reports meeting each		
Risk with interval and horizon	249	249
Reference frame (tier or average risk)	248	249
Phase of care, and why it matters after RT	241	249
Main drivers named	32	249
What the model did not see	0	249
Flags with their reasons (180 reports with a flag)	180	178
Change since the previous scan (123 with a previous scan)	105	123
Input not known on the scan day (8 reports)	0	8
Safety and form		
Forbidden content (diagnosis, RANO category, advice)	0	0
Active flags omitted (of 262)	0	0
Words, median [IQR]	125 [107–135]	190 [158–214]
Reading grade (Flesch–Kincaid), median	11.2	9.7
Reading time at 200 words per minute, median	38 s	57 s

[a] Share of applicable items met. Version 2 − version 1: +0.325 [+0.312, +0.337].

Figure legends

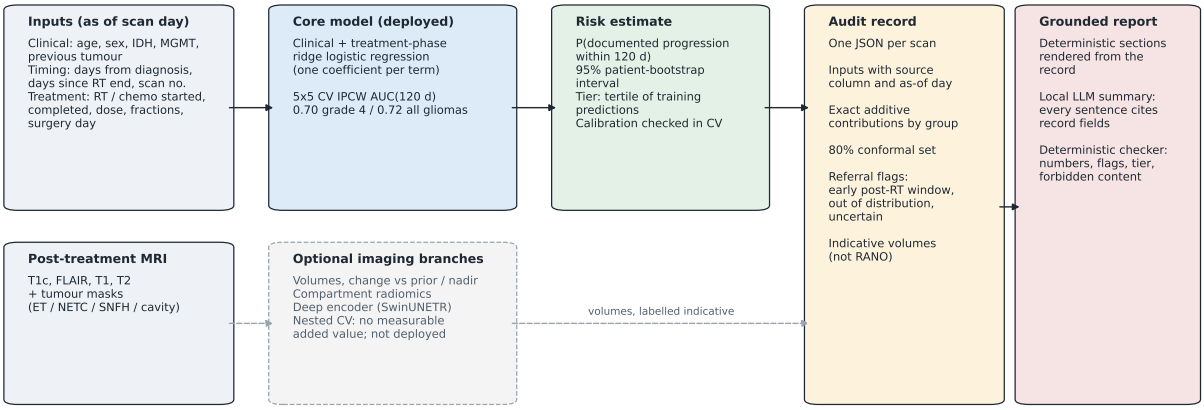


Figure 1. PRISM architecture and data flow. Inputs read as of the scan day feed the deployed core model (ridge on clinical and treatment-course variables). The imaging branches (volumes, radiomics, deep encoder; dashed) are optional and not deployed; volumes enter the record only as indicative values. The risk, with its interval and tier, goes into the per-scan audit record (contributions, 80% conformal set, referral flags), from which the grounded report is written.

[PLACEHOLDER: fig_study_design.pdf]

To be drawn. Three columns, left to right. (1) **Data**: MU-Glioma-Post (155 patients, 308 index scans; grade 4: 126 / 249), LUMIERE (61 / 132, external), CFB-GBM (77 patients with ADC); index-scan rule; label = documented progression within 120 days, censoring at last follow-up or death. (2) **Evaluation**: our initial split (29 test patients) → 500 splits of the same design → 5 × 5 repeated cross-validation grouped by patient (every fitted step inside the training fold; nested selection; stacking); LUMIERE locked external test; CFB-GBM 20 × 5 cross-validation. (3) **Questions**: ablation of the five branches, phase ladder R0–R3 × I1–I4, the five imaging explanations, audit (P1, P2) and reports (v1, v2), each pointing to its rule in Table 4. Vector PDF, colour, same palette and font as Figures 3 to 5.

Figure 2. Study design. The three cohorts, the outcome, the three ways of evaluating the same model and the prespecified questions (rules and answers in Table 4).

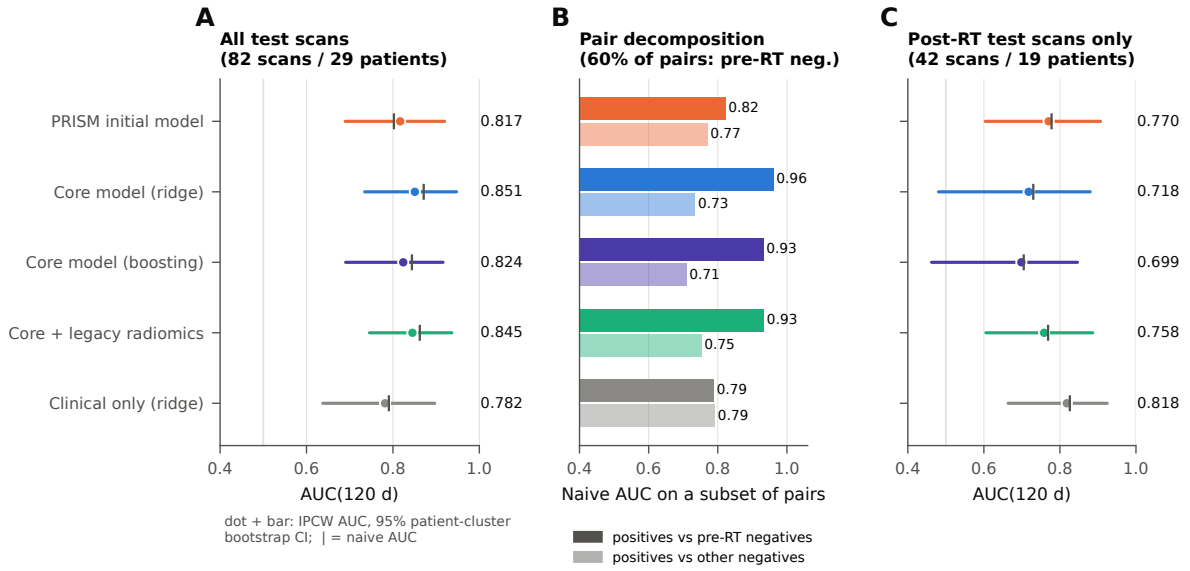


Figure 3. PRISM's initial evaluation split. Discrimination on our initial evaluation split (all gliomas; 126 training / 29 test patients, 82 test scans, 22 naive positives). Every model except PRISM's initial radiomics model was fitted once on the training patients; the initial radiomics model enters with its shipped test-set probabilities. (A) IPCW AUC with 95% bootstrap over patients CI (1,000 draws; dot and bar) and naive AUC (tick). (B) Naive AUC decomposed by the type of negative scan: 36 of the 60 negative test scans were taken before RT started, and none of them progressed, so 60% of positive–negative pairs compare a progression scan with a scan taken before RT; solid bars, positives against negatives taken before RT; light bars, positives against other negatives. (C) The same models on the 42 test scans taken after RT (19 patients). Intervals are wide and overlap throughout.

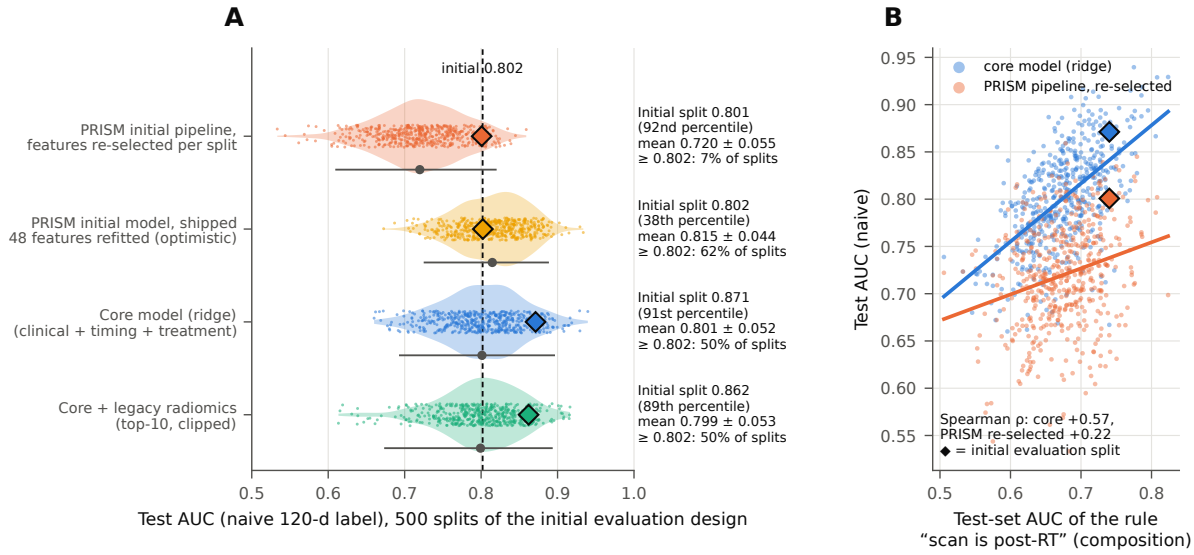


Figure 4. Where our initial evaluation split sits among 500 splits of the same design. PRISM v1's split search was run for seeds 0–499; seed 62 reproduces our initial evaluation split. (A) Test AUC (naive label) per split for PRISM's initial radiomics pipeline with features reselected per split (honest estimate), the shipped features refitted (optimistic upper bound), the core ridge model and the core model + legacy radiomics. Violin and points, 500 splits; grey dot and line, mean and 2.5–97.5% range; diamond, each model's own value on our initial evaluation split with its percentile rank; dashed line, the reported 0.802. (B) Per-split AUC against the AUC of the no-training rule "scan is after RT" on that test set (higher = more negatives and fewer events before RT); lines, least-squares fits; Spearman ρ +0.57 (core model) and +0.22 (initial radiomics pipeline).

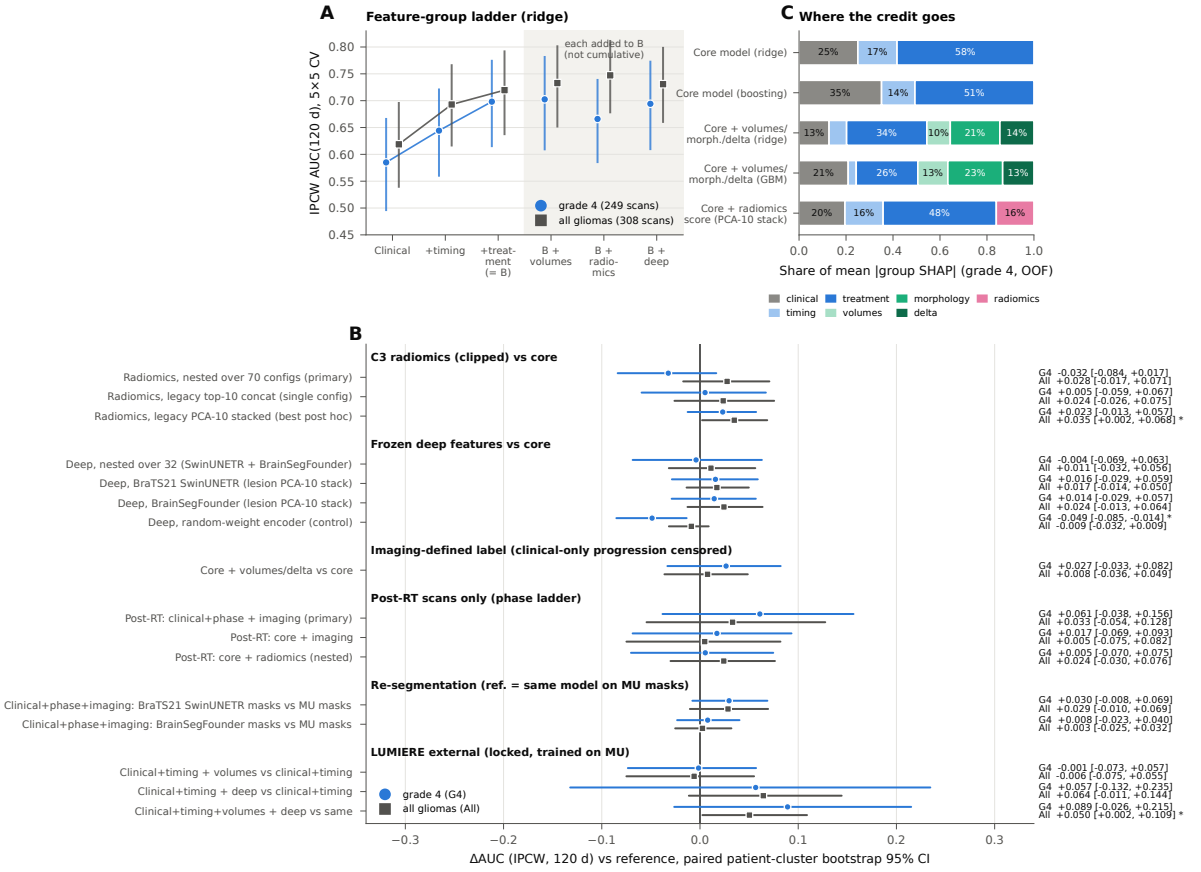


Figure 5. Component ablation: clinical, treatment-phase and imaging branches under repeated cross-validation. (A) Feature ladder (ridge; 5 × 5 repeated CV grouped by patient; IPCW AUC(120 d) with 95% bootstrap over patients CI): clinical → + timing → + treatment status on the scan day (= PRISM's core model, B); the last three points each add one imaging branch to B (volumes; radiomics, nested over 70 configurations; frozen deep features, nested over 32). (B) Paired ΔAUC with 95% CI against the reference named in each section (Table 3). Asterisks mark CIs excluding 0 (3 of 32, uncorrected; about 1.6 expected by chance). (C) Share of mean absolute group SHAP in grade 4 held-out explanations: imaging blocks take 45–51% of the credit in the full models without improving discrimination.

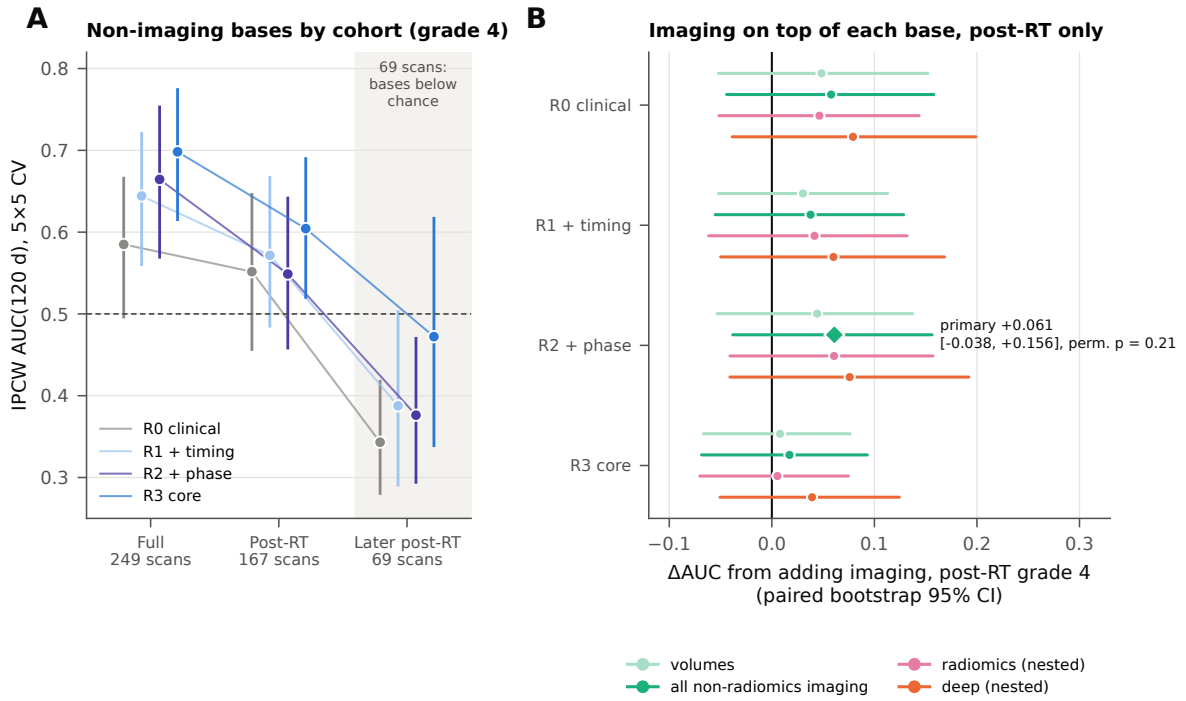


Figure 6. Phase of care. (A) Bases without imaging (ridge, grade 4) in the full cohort (249 scans), the surveillance cohort (scans after RT; 167 scans, 84 patients, 83 events) and the later surveillance cohort (> 12 weeks after RT end; 69 scans), each trained and evaluated within the cohort. R0 clinical; R1 + timing; R2 + four-level phase of care; R3 core model. In the later surveillance cohort the bases fall below 0.5, as expected for a nearly null model in a small sample, so contrasts there are not interpretable. (B) ΔAUC from adding each imaging block (I1 volumes; I2 all imaging measurements other than radiomics; I3 radiomics; I4 deep) to each base within grade 4 surveillance scans. Diamond, prespecified primary contrast (R2 + I2 vs R2): +0.061 (−0.038 to +0.156), permutation p = 0.21; the permutation null itself averages +0.03.

A

PRISM surveillance report

PatientID_0003_Timepoint_1 | scan day 90 from diagnosis | early post-RT (0-84 days after the end of radiotherapy)

Risk of documented progression within 120 days

0.46 (95% interval 0.26-0.71)

Tier: middle third of development-cohort predictions (cut points 0.26 / 0.50). 80% conformal set: both outcomes (progression and no progression).

Referral flags

[!] early post-RT window: scan 25 days after the end of RT, inside the 84-day early post-RT window where treatment effects can mimic progression

[!] out of distribution: MGMT status unknown

[!] uncertain: the 80% conformal prediction set contains both outcomes

Action: flagged for clinician review

Contributions (exact additive log-odds; positive = towards higher risk)

Intercept -0.39 clinical +0.14 timing -0.34 treatment +0.43 = logit -0.17. Imaging: not part of this model.

-0.27 MGMT status unknown: yes (clinical)
-0.27 scan number = 1 (timing)
+0.16 RT started: yes (treatment)
+0.16 male sex: no (clinical)
+0.15 previous brain tumour: no (treatment)

Inputs as of the scan day (source: MU clinical workbook)

Age 57, female; IDH wildtype; MGMT unknown; RT started yes, completed yes; days since RT end 25; chemotherapy started yes. All 23 inputs known by the scan day: yes (each with source column and as-of day in the record).

Volumetric change (indicative, not RANO)

Enhancing 33.8 cc; FLAIR hyperintensity 44.3 cc.

First scan: no prior or nadir.

Summary (local LLM; every sentence cites record fields; checked deterministically)

FINDINGS: This scan is 90 days from diagnosis, 25 days after radiotherapy ended, in the early post-RT phase. In this window, treatment-related change can look like progression on MRI, which the model cannot separate. The model estimates a 46% (95% interval 26-71%) risk of documented progression within 120 days. This is in the middle third of the model's estimates on its training scans, above the average. The largest contributions are MGMT status unknown and scan number one, both towards lower risk. The model did not use the MRI images or tumour volumes, steroid dose, or symptoms. The scan is flagged for clinician review due to the early post-RT window, out of distribution, and uncertain status.

IMPRESSION: The model estimates a 46% risk, in the middle third above the average, with active flags for clinician review.

Faithfulness check: 8/8 claims supported; active flags mentioned: all; forbidden content: none.

Limitations

Model discrimination in development cross-validation AUC 0.70 (single institution, MU-Glioma-Post); the risk is the probability of documented progression within 120 days, not a diagnosis of progression or pseudoprogression; volumes are indicative, not a RANO assessment.

B

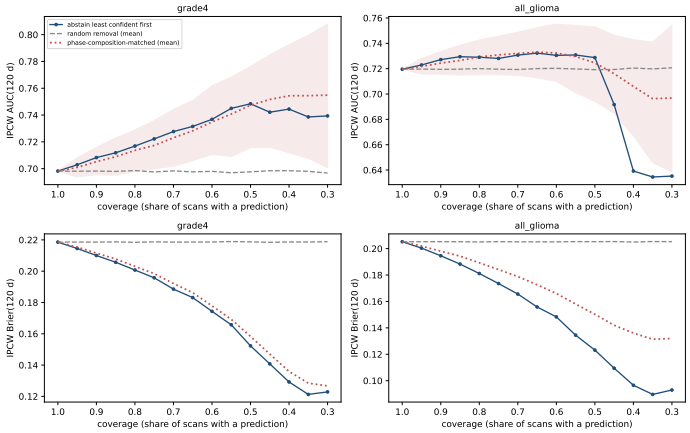


Figure 7. A grounded report and selective prediction. (A) Grounded report (version 2) for the first evaluation scan in the 12 weeks after RT, chosen by a prespecified rule: sections rendered from the audit record and a local language model summary, with findings and an impression, in which every sentence cites record fields. The checker found every claim supported. By the usefulness rule this is one of the 2 of 249 incomplete reports: it names the active flags without their reasons. (B) Risk–coverage curves for the core model (5×5 CV, pooled held-out predictions, mean over repeats), grade 4 (left) and all gliomas (right). Top, IPCW AUC(120 d); bottom, IPCW Brier score of the scans kept when the least confident scans (lowest $\max(p, 1-p)$) are withheld first (solid), against random removal (dashed) and random removal matched by phase (dotted; band, 2.5–97.5% of the matched draws). The gain over random removal is almost entirely matched by the phase-matched reference (prespecified P2 at about 70% coverage: +0.006, −0.023 to +0.036, grade 4).